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Dissecting tumor transcriptional heterogeneity from single-cell RNA-seq data by generalized binary covariance decomposition

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CONTENTS

Supplementary note 1
References 6
Supplementary figures 9

Supplementary note

GBCD model fitting procedure. In this section we describe how to fit the EBMF model (6) with a GB prior (5). Fitting an EBMF model reduces to solving a series of simpler empirical Bayes normal means (EBNM) problems [1]. Therefore, here we describe how we solve the EBNM problem for a normal means model with a GB prior.

Let $\boldsymbol{x} = (x_1, \ldots, x_n)^T$ be the vector of observed means, let $\boldsymbol{s} = (s_i, \ldots, s_n)^T$ be the vector of corresponding standard errors, and let $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)^T$ be the vector of unknown means to be estimated. The normal means model with GB prior is

$$x_i \mid \theta_i \overset{\text{ind.}}{\sim} N(\theta_i, s_i^2), \tag{16}$$

$$\theta_i \overset{\text{i.i.d.}}{\sim} (1-\pi)\delta_0 + \pi N_+(\mu, \sigma^2). \tag{17}$$

The hyperparameters μ, σ^2, π are estimated by maximizing the likelihood as we describe next.

The marginal likelihood is

$$\begin{aligned} L(\mu, \sigma^2, \pi; \boldsymbol{x}, \boldsymbol{s}) &= \prod_{i=1}^n \left[(1-\pi)N(x_i; 0, s_i^2) + \pi \textstyle\int N(x_i; \theta_i, s_i^2) N_+(\theta_i; \mu, \sigma^2)\, d\theta_i\right] \\ &= \prod_{i=1}^n \left[(1-\pi)N(x_i; 0, s_i^2) + \pi N(x_i; \mu, \sigma^2 + s_i^2)\, \frac{\Phi(\tilde{\mu}_i/\tilde{\sigma}_i)}{\Phi(\mu/\sigma)}\right], \end{aligned} \tag{18}$$

where $N(\,\cdot\,; \mu, \sigma)$ and $N_+(\,\cdot\,; \mu, \sigma)$ denote the probability density function of $N(\mu, \sigma^2)$ and $N_+(\mu, \sigma)$, respectively, $\Phi(\,\cdot\,)$ denotes the cumulative distribution function of the standard normal distribution, and we define

$$\tilde{\sigma}_i^2 := \frac{1}{1/\sigma^2 + 1/s_i^2}, \tag{19}$$

$$\tilde{\mu}_i := \frac{s_i^2 \mu + \sigma^2 x_i}{s_i^2 + \sigma^2}. \tag{20}$$

The ratio σ/μ is assumed to be fixed to some (typically small) value, ω, so the maximization problem reduces to two parameters μ and π, and we set $\sigma = \omega\mu$.

To solve this maximization problem, we first define the following data augmentation. For each i, we define latent variable $z_i \in \{0, 1\}$ that is 1 if θ_i is drawn from $N_+(\mu, \sigma^2)$, and $z_i = 0$ otherwise. Marginalizing over z_i recovers the original model. This data augmentation leads to the following EM algorithm [2]. The E-step calculates the posterior expectation of z_i given the current hyperparameter estimates $\mu^{(t)}$ and $\pi^{(t)}$,

$$\zeta_i^{(t)} := \mathbb{E}[z_i \mid x_i, s_i, \mu^{(t)}, \pi^{(t)}], \quad i = 1, \ldots, n. \tag{21}$$

The M-step update for π has the following closed-form solution:

$$\pi^{(t+1)} = \operatorname*{argmax}_{\pi \in [0,1]} \sum_{i=1}^n \left\{(1-\zeta_i^{(t)})\log(1-\pi) + \zeta_i^{(t)} \log \pi\right\} = \frac{1}{n}\sum_{i=1}^n \zeta_i^{(t)}. \tag{22}$$

The M-step update for μ is

$$\mu^{(t+1)} = \underset{\mu \geq 0}{\operatorname{argmax}} \sum_{i=1}^{n} \zeta_i^{(t)} \left\{ \log N(x_i; \mu, \omega^2\mu^2 + s_i^2) + \log \Phi(\tilde{\mu}_i/\tilde{\sigma}_i) \right\}. \tag{23}$$

This does not have a closed-form solution, so we used the "L-BFGS-B" numerical optimization method available via the R function `optim` to solve for $\mu^{(t+1)}$. Both M-step updates maximize the expected value of the complete log-likelihood. The complete log-likelihood is

$$\begin{aligned} \ell(\mu, \omega, \pi; \boldsymbol{x}, \boldsymbol{s}, \boldsymbol{z}) = \sum_{i=1}^{n} \Bigg[& (1 - z_i) \log \left((1 - \pi)\, N(x_i; 0, s_i^2) \right) \\ & + z_i \log \left(\pi\, N(x_i; \mu, \omega^2\mu^2 + s_i^2)\, \frac{\Phi(\tilde{\mu}_i/\tilde{\sigma}_i)}{\Phi(1/\omega)} \right) \Bigg], \end{aligned} \tag{24}$$

The E-step and M-step alternate at each iteration $t = 1, 2, \ldots$ until convergence. We denote the converged estimates of μ, π, ζ_i by $\hat{\mu}, \hat{\pi}, \hat{\zeta}_i$. We then compute posterior expectations of the unknown means θ_i $(i = 1, \ldots, n)$. The posterior distribution of θ_i given $\hat{\mu}, \hat{\pi}$ is

$$p(\theta_i \mid x_i, s_i, \hat{\mu}, \hat{\pi}) = (1 - \hat{\zeta}_i)\delta_0 + \hat{\zeta}_i N_+(\hat{\tilde{\mu}}_i, \hat{\tilde{\sigma}}_i^2), \tag{25}$$

where

$$\hat{\tilde{\mu}}_i = \frac{s_i^2 \hat{\mu} + \omega^2 \hat{\mu}^2 x_i}{s_i^2 + \omega^2 \hat{\mu}^2}, \tag{26}$$

$$\hat{\tilde{\sigma}}_i^2 = \frac{1}{1/(\omega^2\hat{\mu}^2) + 1/s_i^2}. \tag{27}$$

Having described and implemented the EBNM solver for a normal means model with a GB prior, we can fit a GBCD model by providing this EBNM solver to the function `flash` in the flashier R package [3] (via the `ebnm.fn` argument). In detail, fitting the "relaxed model" (6) consists of four seps in which $\boldsymbol{L}$ and $\tilde{\boldsymbol{L}}$ are initialized in the first three steps, then these initial estimates are refined in the fourth step. Steps 1 through 4 are as follows:

(i) Add up to $K_{\max}$ pairs of $\boldsymbol{l}_k, \tilde{\boldsymbol{l}}_k$ using function `flash_greedy`, in which Laplace priors are assigned to the elements of $\boldsymbol{l}_k$ and $\tilde{\boldsymbol{l}}_k$.

(ii) For each k, improve estimates of $\boldsymbol{l}_k, \tilde{\boldsymbol{l}}_k$ using function `flash_backfit`. This function iterates until the convergence criterion is met, or until it reaches T_1 iterations, whichever comes first.

(iii) For each k, split $\boldsymbol{l}_k$ into two nonnegative vectors $\boldsymbol{l}_k^+$ and $\boldsymbol{l}_k^-$, where $l_{ik}^+ = \max\{0, l_{ik}\}$ and $l_{ik}^- = \max\{0, -l_{ik}\}$, so that $\boldsymbol{l}_k = \boldsymbol{l}_k^+ - \boldsymbol{l}_k^-$. Do the same for $\tilde{\boldsymbol{l}}_k$ to obtain $\tilde{\boldsymbol{l}}_k^+$ and $\tilde{\boldsymbol{l}}_k^-$.

(iv) For each k, improve the estimates of $\boldsymbol{l}_k^+, \tilde{\boldsymbol{l}}_k^+, \boldsymbol{l}_k^-, \tilde{\boldsymbol{l}}_k^-$ using function `flash_backfit`. GB priors are assigned to all elements of $\boldsymbol{l}_k^+$, $\tilde{\boldsymbol{l}}_k^+$, $\boldsymbol{l}_k^-$, $\tilde{\boldsymbol{l}}_k^-$. This function iterates until the convergence criterion is met, or until it reaches T_2 iterations, whichever comes first.

For all our final analyses of the simulated and real datasets, we set $T_1 = 500$ and $T_2 = 200$. In the simulations, running these steps for a single dataset (3,200 cells, 10,000 genes), with $T_1 = 500$ and $T_2 = 200$, took about 30 minutes and used at most 2 GB of memory.

Optimization of the GBCD pipeline. As we explained in the Methods ("Computing requirements"), our original GBCD implementation performed computations on the (non-sparse) $N \times N$ matrix $\boldsymbol{Y}\boldsymbol{Y}^T$, and was used to generate all the results presented in the paper.

However, this implementation scales poorly with N, and therefore is not well suited for large datasets. We subsequently improved the GBCD implementation in the gbcd R package so that it can perform computations using the (sparse) $N \times J$ matrix $\boldsymbol{Y}$ (without forming $\boldsymbol{Y}\boldsymbol{Y}^T$). This can achieve significant reductions in runtime and memory usage when $\boldsymbol{Y}$ is sparse and N is large.

To verify that the two implementations produce similar results, we ran GBCD on the PDAC dataset. This dataset is a good test case because it is a large and complex dataset. The original implementation (that forms $\boldsymbol{Y}\boldsymbol{Y}^T$) completed the model fitting in about 50 h and used as much as 200 GB of memory, whereas the new implementation (that does not form $\boldsymbol{Y}\boldsymbol{Y}^T$) completed in about 21 h and used at most 48 GB of memory. Overall, the results from the two implementations were very consistent (Supplementary Fig. 17).

We also examined whether the running time could be decreased by reducing the number of backfitting iterations; the most time-consuming steps in GBCD are the steps that involve running T_1 and T_2 backfitting iterations. We again used the PDAC dataset to examine this.

Supplementary Fig. 17 shows the time spent on backfitting as a function of the number of iterations. The runtime per backfiting iteration in step (iv) was much longer than that in step (ii). This was likely due to the fact that there were more factors to update in each iteration in step (iv) than in step (ii). While the choice of T_1 strongly affected a few GEPs (e.g., GEP2), the estimates of all GEPs stabilized at a T_2 setting of 25. Although it is difficult to draw general conclusions from this experiment, it suggests that GBCD results are less sensitive to T_2, and it might be reasonable to set T_2 to a small number (e.g., 25) to reduce running time when N is large. (Note that the results presented in the paper were obtained with $T_1 = 500$ and $T_2 = 200$). When we combined the new GBCD implementation with fewer backfitting iterations ($T_1 = 500$, $T_2 = 25$), this further lowered the running time from 21 h to 14 h, and produced very similar results.

Additional details on the methods compared: matrix factorization methods. Here we describe in detail how we ran the other matrix factorization methods. A subset of these methods were also applied to the HNSCC and PDAC datasets. In these descriptions, $\boldsymbol{X}$ denotes the UMI counts matrix and $\boldsymbol{Y}$ denotes the normalized data matrix. In all cases, the elements of $\boldsymbol{Y}$ can be interpreted as "shifted log counts" (see "Preparation of scRNA-seq data" in Methods).

PCA. We computed principal components from $\boldsymbol{Y}$ using the R function `svd`.

Patient-by-patient NMF. This method involved fitting an NMF separately to the UMI counts from each patient. Each NMF was fit to the data by calling the `nnmf` function from the R package NNLM [4]. Each NMF was fit by optimizing the mean-squared loss function (which is the default setting in NNLM), with $K = 3$. We set `method = "scd"` and `rel.tol = 1e-8`.

Following [5], we manually identified clusters of highly correlated programs ("meta-programs"). The similarity between a pair of programs was calculated using the fraction of genes shared among the top-ranked 50 genes (as quantified by the Jaccard index). The memberships for each meta-program were defined as follows: if a factor from a given patient belonged to the meta-program cluster, the membership of cells from that patient was set to the NMF membership estimate for that factor; otherwise, the membership of cells from that patient was zero.

Combined NMF (UMI counts). We fit a Poisson NMF model to $\boldsymbol{X}$, which is equivalent to fitting a multinomial topic model [6], by calling the `fit_topic_model` function from the R package fastTopics. We set `numiter.main = 100` and `numiter.refine = 500`, and kept all other options at their default settings.

Combined NMF (shifted log counts). We fit an NMF to $\boldsymbol{Y}$ by calling the `nnmf` function from the R package NNLM [4]. The NMF was fit by optimizing the mean-squared loss function (which is the default setting in NNLM). We set `method = "scd"` and `rel.tol = 1e-8`, and kept all other options at their default settings.

Consensus NMF. We ran consensus NMF [7] on $\boldsymbol{X}$ using the Python package cNMF. In the HNSCC and PDAC datasets, we ran consensus NMF on the normalized counts $\boldsymbol{Y}$.

LIGER. We ran LIGER [8] on the UMI counts $\boldsymbol{X}$ using the R package rliger. For the HNSCC data, the UMI counts were not available, so we ran LIGER on $\boldsymbol{Y}$. We followed the recommended analysis pipeline described in the package vignette. In the simulations, we also assessed the impact of the regularization parameter, λ, on LIGER's performance (Supplementary Fig. 4). The LIGER results shown in Fig. 1 were obtained with $\lambda = 5$. We used the `H.norm` output, the normalized cell loading estimates, to evaluate LIGER.

CD, point-exp prior. This was the same method as GBCD, except that the GB priors were replaced with point-exponential priors [9]. As with GBCD, the model was fit to $\boldsymbol{Y}$.

EB-SNMF (point-exp prior or GB prior). These methods involved fitting EBMF models to $\boldsymbol{Y}$ using flashier, in which point-Laplace priors (9) were assigned to $\boldsymbol{F}$, and GB priors (5) or point-exponential priors were assigned to$\boldsymbol{L}$. Unlike GBCD, no orthogonality constraint was enforced on $\boldsymbol{F}$. Fitting an EB-SNMF model involved the following four steps:

(i) Add up to $K_{\max}$ pairs $\boldsymbol{l}_k$, $\boldsymbol{f}_k$ using function `flash_greedy`, in which point-Laplace priors are assigned to the elements of $\boldsymbol{l}_k, \boldsymbol{f}_k$.

(ii) For each k, improve estimates of $\boldsymbol{l}_k$ and $\boldsymbol{f}_k$ using function `flash_backfit`. This function iterates until the convergence criterion is met, or until it reaches T_1 iterations, whichever comes first.

(iii) For each k, split each $\boldsymbol{l}_k$ into two nonnegative vectors $\boldsymbol{l}_k^+$ and $\boldsymbol{l}_k^-$, where $l_{ik}^+ = \max\{l_{ik}, 0\}$ and $l_{ik}^- = \max\{-l_{ik}, 0\}$, so that $\boldsymbol{l}_k = \boldsymbol{l}_k^+ - \boldsymbol{l}_k^-$; set $\boldsymbol{f}_k^+ = \boldsymbol{f}_k$ and $\boldsymbol{f}_k^- = -\boldsymbol{f}_k$.

(iv) For each k, improve the estimates of $\boldsymbol{l}_k^+, \boldsymbol{f}_k^+, \boldsymbol{l}_k^-, \boldsymbol{f}_k^-$ using function `flash_backfit`. GB or point-exponential priors are assigned to all elements of $\boldsymbol{l}_k^+, \boldsymbol{l}_k^-$ and point-Laplace priors are assigned to all elements of $\boldsymbol{f}_k^+, \boldsymbol{f}_k^-$. This function iterates until the convergence criterion is met, or until it reaches T_2 iterations, whichever comes first.

Additional details on the methods compared: batch-correction methods. In this section we describe in detail how we ran the batch-correction methods. A subset of these methods were also applied to the HNSCC data.

For CCA, Conos and MNN Correct, we ran NMF on the "batch-corrected" data in the same way that we ran "combined NMF" on the shifted log counts (see above). In the simulations, we ran NMF with $K = 12$. Then we used the NMF to evaluate performance of the method; that is, after running NMF, we compared the estimated memberships from the NMF (the $\boldsymbol{L}$ matrix) to the true memberships.

CCA (Seurat). We ran CCA on the shifted log counts following the standard integration workflow described in one of the Seurat package vignettes [10]. This involved the following steps. First, for each patient, we identified the 5,000 most variable genes from the scRNA-seq data for that patient (this step was implemented using the `FindVariableFeatures` function from Seurat). Then we found a set of "shared anchors" across patients using the `FindIntegrationAnchors` function, with `anchor.features = 5000`, `dims = 1:50`. Finally, we performed a data integration across patients based on these shared anchors using the `IntegrateData` function, with `dims = 1:50`.

MNN Correct. We applied MNN Correct to the shifted log counts by calling the `mnnCorrect` function in the batchelor R package [11] .

Conos. We ran Conos on the shifted log counts using the conos R package [12]. For each patient, we kept the scRNA-seq data for the 8,000 most variable genes in that patient (this step was implemented using the `FindVariableFeatures` function from Seurat). Then we constructed a Conos object using the `Conos$new` function, and built the joint graph using the `buildGraph` function with `k = 15`, `k.self = 5`, `n.odgenes = 5000`. Finally, we computed batch-corrected gene expression data using the `correctGenes` function, with `n.od.genes = 8000`.

Choice of K in the simulations. Determining the number of factors, K, in a matrix factorization is generally a challenging task. For NMF fit to the UMI counts, we set $K = 11$, the same setting used to simulate the data. For NMF fit to the shifted log counts, we set $K = 12$ to allow for an additional factor that captures baseline gene expression levels. For consensus NMF, we tried several settings of K ranging from 8 to 12, and chose a setting K that achieved a good balance of stability and accuracy. For PCA, we tried several settings of K, $K = 8, 10, 12$, to assess the impact of the number of PCs on performance.

All the EBMF methods we compared (including GBCD) automatically select K (up to the user-specified $K_{\max}$). In the simulations, we set $K_{\max} = 16$ for all the EBMF methods.

Additional discussion of the PDAC study and results.

Background on PDAC molecular subtyping. Many studies have attempted to define clinically relevant subtypes with the goal of personalizing treatment and improving outcomes [13]. An early study applied NMF to microarray expression data and identified "classical" and "basal" subtypes [14]. Later studies used newer technologies to refine these subtypes: Chan-Seng-Yue et al. [15] identified 4 subtypes — "classical-A", "classical-B", "basal-A" and "basal-B" — from an NMF analysis of bulk RNA-seq data in tumor specimens that underwent laser capture microdissection to improve tumor purity; Raghavan et al. [16] refined the basal and classical gene signatures from [14] using scRNA-seq data from metastatic patients; and Hwang et al. [17] identified one classical and three basal-related expression programs ("basaloid", "squamoid" and "mesenchymal") from an NMF analysis of single-nucleus RNA-seq data in primary tumors.

Biological interpretation of GEP14. Enrichment analysis of the driving genes of GEP14 highlights several gene sets related to stress response (Supplementary Fig. 14). These include genes upregulated in response to hypoxia (e.g., *BNIP3L, DDIT3, VEGFA*). Hypoxia gene expression signatures have been previously connected to worse PDAC outcomes, and hypoxia is a promising target of innovative therapeutic strategies [18–22]. Other notable driving genes related to stress response include activating transcription factor 4, *ATF4* — the main effector of the integrated stress response (ISR) [23] — and several of its validated targets, including *DDIT3/CHOP, HSPA5, MAP1LC3B, MTHFD2, SQSTM1* [24–26]. Among these, *SQSTM1* and *MAP1LC3B* are key modulators of autophagy [25, 27], which is known to promote PDAC progression and therapy resistance [28, 29], and *HSPA5* has been previously linked with PDAC progression through decreased ferroptosis [30, 31].

ISR is activated in response to stress, including hypoxia, by phosphorylation of the α subunit of eukaryotic initiation factor 2 (eIF2α) at serine 51, which suppresses global protein synthesis to reduce energy expenditure while enhancing translation of mRNAs of specific genes, including *ATF4*. This then upregulates the expression of many genes involved in stress

adaptation [26]. Recently, *ATF4* has been shown to promote amino acid biosynthesis and tumor cell survival in PDAC [32]. Also, eIF2α phosphorylation significantly correlates with tumor recurrence/metastasis and shorter overall survival [33]. Interestingly, the *ATF4* expression signature has been found to be a predictor of therapy resistance in other cancers. For example, it was revealed to be induced in melanoma cells that escaped RAF kinase inhibitor treatment in a recent scRNA-seq study [34]; tumor cells that persisted after treatment with BH3 mimetics similarly depended on *ATF4* and the ISR to survive [35]. The fact that GEP14 includes many genes induced during the ISR, and is an independent predictor of PDAC patient survival, suggests an important role for the ISR in PDAC progression, and raises the potential that pharmacological interventions targeting the ISR could provide useful therapeutic strategies. More work is needed to test these hypotheses.

Results of alternative matrix factorization methods. The results from applying PCA and consensus NMF to the PDAC data highlighted behaviors qualitatively similar to what we observed in the simulations and the HNSCC data (Supplementary Figures 18 and 19). Consensus NMF identified many patient-specific programs but failed to identify classical- or basal-related programs. PCA failed to identify patient-specific effects as individual PCs, and instead captured them using combinations of PCs.

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